

Visible-to-Infrared Image Translation

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Abstract. Visible to thermal image translation is a significant part of research for applications needing robust perception under challenging visual conditions, like surveillance, autonomous navigation, and night vision. Generating reliable thermal representations from RGB images is a difficult task, and many datasets in the visible domain require their thermal counterparts for many applications. This work presents a comparative study of three generative adversarial models, Pix2Pix, InfraGAN, and ClawGAN, implemented and scaled down to match resource availability. The study evaluates the outputs of these models through a combination of metrics (PSNR, SSIM, LPIPS, and FID) in addition to qualitative analysis. Overall, this study offers insight into the relative strengths of modern GAN-based methods for thermal image synthesis. We investigate three GAN-based approaches—Pix2Pix (paired baseline) [1], a reconstructed InfraGAN (structurally supervised IR synthesis) [2], and our own CLAWGAN (context-aware thermal generation with attention) [3]. All models are trained under strict hardware constraints (single 6 GB GPU) for a fair comparison. We document architectures, mathematical loss formulations, hyperparameters, and outcomes. Quantitative metrics (PSNR, SSIM, LPIPS, FID) and qualitative visual analyses are provided.

Keywords: GANs, Image-to-Image Translation, Thermal Imaging, Pix2Pix, InfraGAN, CLAWGAN, Structural Losses, Attention

1 Introduction

Infrared imaging is integral to safety-critical applications like target detection, autonomous driving, and surveillance systems. They allow perception under low visible illumination. Despite this, infrared imaging is limited because infrared cameras capture noise and may have low spatial resolution compared to their visible-spectrum counterparts. Environmental factors such as heat reflections and background emissions can further degrade infrared quality. Moreover, many applications require visible-infrared image pairs to train models for generative and detection purposes, creating a need for alternatives beyond traditional thermal cameras.

Recent advances in image-to-image translation, particularly those driven by Generative Adversarial Networks (GANs), have shown significant potential in

learning complex cross-modal mappings. GAN-based techniques have demonstrated exceptional performance in style transfer, data augmentation, and modality transformation, making them viable options for visible-to-thermal translation. By using these deep learning methods, thermal properties can be estimated from visual inputs, reducing reliance on thermal sensors and increasing the applicability of thermal imaging.

The central objective of this work is to convert visible-spectrum images into corresponding high-quality thermal representations using deep generative models. Unlike traditional image enhancement or style transfer tasks, visible-to-thermal translation involves learning a fundamentally nonlinear mapping between two distinct physical domains. Thermal images encode heat emission rather than reflected light, meaning that surface temperature, material properties, and environmental factors must be inferred from visual cues alone, making the translation problem inherently ambiguous and challenging.

2 Problem Definition

The main goal of this paper is to convert visible-spectrum images into corresponding infrared images using generative adversarial networks. Unlike traditional image enhancement or style transfer tasks, visible-to-thermal translation involves learning a fundamentally nonlinear mapping between two distinct physical domains. Surface temperature, material characteristics, and environmental elements must be deduced solely from visual cues since infrared images record heat emission rather than reflected light. Preserving structural features while mitigating texture loss during generation is a key challenge. A robust model must capture these interactions, maintain spatial consistency, and produce outputs that closely mimic real infrared data.

3 Objectives

This work aims to evaluate and compare three models for visible-to-infrared image translation; Pix2Pix[4] as a baseline paired translation method, InfraGAN[2] with structural IR reconstruction, and CLAWGAN[3] for context-aware thermal synthesis. Each model is assessed both quantitatively and qualitatively. The study also identifies the strengths, limitations, and practical suitability of these methods under constrained hardware resources.

4 Related Work

InfraGAN (2022) proposed a conditional GAN for visible-to-infrared translation, using SSIM and pixel-based losses to preserve both structural and appearance fidelity. It was evaluated on the KAIST[5] and VEDAI[6] datasets. The model produces visually plausible IR images and preserves structure well, though it relies on paired data and its quality is tied to dataset diversity [2].

ThermalGAN (2018/2019) was an early attempt at color-to-thermal translation, targeting person and object re-identification. Using the ThermalWorld dataset, it demonstrated the feasibility of multimodal translation, but was limited in scene diversity and sometimes generated oversimplified IR outputs [7].

IR-GAN (2023) improved conditional GANs by employing a ConvNeXt-based generator and multi-scale PatchGAN discriminator, along with a gradient-based loss to enhance edges and textures. Evaluated on public datasets like IVFG, it produces sharper IR images with better structural fidelity, but comes with higher computational cost and potentially lower generalization across varied scenes [8].

TransImg (2024) combined a deep residual encoder with region-perception feature fusion for better detail and generalization. Tested on a large-scale paired dataset of over 63k image pairs, it showed improved detail preservation and noise control compared to conventional GANs. However, the model is large, requires longer training, and still treats IR more as a “style” than physically accurate thermal data [9].

MWIRGAN (2023) is an unsupervised CycleGAN-style method for visible-to-MWIR translation, enhanced with a perceptual loss to better preserve object shapes. It does not require paired images and can also use synthetic data for augmentation. Its weaknesses include less precise mapping, potential inaccuracies in thermal properties, and reduced performance in complex scenes [10].

TM-GAN integrates visible-to-IR translation with downstream template-matching tasks using a composite loss. It ensures generated IR images are structurally useful for matching, performing better than baseline methods in both image quality and matching accuracy. The method is task-specific and relies on paired data, which may limit general applicability [11].

IRKDGAN (2025) uses discriminator-guided knowledge distillation with multi-scale fusion to improve convergence and generate finer IR details. Evaluated on large benchmark datasets such as FLIR, KAIST, and AVIID, it improves IR quality over prior GANs but still requires paired data and substantial computational resources [12].

PID (Physics-Informed Diffusion, 2025/2026) applies latent diffusion models with physics-based constraints (temperature, emissivity, thermal texture) to generate physically consistent IR images. Using KAIST and FLIR datasets, it improves alignment with real IR data but involves complex training, slower inference, and requires IR data for supervision [4].

IRFormer (2024) is a Transformer-based visible-to-IR model, combining color perception, dynamic feature fusion, and attention mechanisms to better capture global context and preserve details. Tested on LLVIP, RoadScene, M3FD, FLIR, and MCube datasets, it generates high-fidelity IR images, though it is computationally heavy and limited evaluation may affect generalization [13].

DiffV2IR (2025) uses a multi-stage diffusion-based approach combined with vision-language guidance to achieve semantic-aware, structure-preserving visible-to-IR translation. Trained on the large IR-500K dataset, it handles diverse IR wavelength distributions and avoids common GAN issues like mode collapse. Its

limitations include heavy computation, large dataset requirements, and outputs that may be visually plausible but not fully physically accurate [14].

5 Methodology

5.1 Models Under Evaluation

Pix2Pix[1] is a general-purpose conditional GAN that establishes a contemporary baseline for supervised translation tasks like colorization and label-to-image synthesis by learning a mapping from input to output using an adversarial loss.

Using an encoder-decoder generator, a pixel-wise discriminator, and an SSIM-based structural loss to manage dataset scarcity and enhance cross-modal geometry preservation, InfraGAN[2] is intended for visible-to-infrared translation.

In order to achieve better visual quality and higher recognition accuracy, particularly with unregistered pairs, ClawGAN[3] uses a CycleGAN-inspired architecture with a "claw-connected" generator, mismatch metrics, and template-matching alignment for thermal-to-visible facial image translation.

5.2 Resource and Computational Constraints

Due to hardware limitations, we downscaled both the dataset and model architectures to ensure feasible training and evaluation on our system. We used a subset of RGB–Thermal pairs instead of the full KAIST[5] dataset, and adjusted network parameters to reduce memory usage while maintaining performance.

All experiments were conducted on a workstation equipped with an AMD Ryzen 7 CPU, an NVIDIA RTX 4060 GPU, and 24 GB of RAM. Training times varied depending on the model complexity, ranging from several hours for lightweight models to roughly one day for larger architectures. These adjustments allowed us to conduct all experiments efficiently without compromising the integrity of the comparative study.

5.3 Dataset Description

KAIST Multispectral Pedestrian Dataset We make use of the 95,000 RGB–Thermal picture pairs (resolution 640×480 , 20 Hz) from a moving vehicle that make up the KAIST Multispectral Pedestrian Dataset [5]. Pedestrians, persons, and bicycles are carefully tagged for each pair, yielding 103,128 dense annotations and 1,182 distinct pedestrian instances. We choose a subset of 20,000 picture pairs for this investigation, splitting them into an 80–20 train/test split. The dataset’s variety of scenes daytime, nighttime, indoor, and outdoor—ensures a reliable assessment of thermal image synthesis under various circumstances.

Each RGB–Thermal pair is preprocessed to ensure consistency and alignment between modalities. Both RGB and thermal images are center-cropped to a fixed size of 512×512 pixels, maintaining uniform input dimensions for the

models. RGB images are normalized per channel to the range $[-1, 1]$, which helps stabilize GAN training, while thermal (LWIR) images are converted to grayscale and normalized to the same range. Only pairs where both RGB and thermal images exist are included, guaranteeing precise alignment for supervised learning.



Fig. 1: A visible (RGB) and corresponding thermal (LWIR) image pair from the KAIST dataset.

IRIS: Thermal-Visible Face Images We additionally used IRIS: Thermal-Visible Face Images dataset [15] of simultaneously acquired thermal and visible face images under variable illuminations, expressions, and poses. Thermal images were captured using a Raytheon Palm-IR-Pro sensor, and visible images using a Panasonic WV-CP234 camera. The dataset contains 4,228 image pairs from 30 individuals, with 176–250 images per person, covering three expressions (surprised, laughing, angry), multiple poses, and six illumination conditions (left, right, both, dark, off, varying). Each image is sized 320×240 pixels, and the total dataset size is approximately 1.83 GB.

Due to slight misalignment between the thermal and visible sensors, we employed a template-matching-based alignment strategy. Specifically, a cropped template from the thermal image was matched against the visible image in grayscale to find the best correspondence, allowing precise alignment while preserving RGB information for the network. Following alignment, both thermal and visible images were resized to a fixed training resolution and normalized to the range $[-1, 1]$. Data augmentation included random horizontal flips during training to improve generalization.

This preprocessing ensures that visible-to-thermal pairs are properly registered and suitable for supervised learning, particularly for models that rely on pixel-to-pixel correspondence such as CLAWGAN[3].

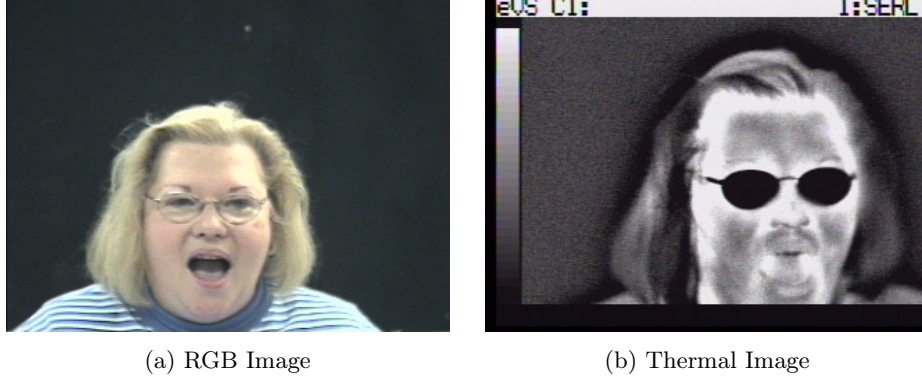


Fig. 2: A visible (RGB) and corresponding thermal (LWIR) image pair from the IRIS dataset.

Data preprocessing In order to prepare both visual modalities for the neural network, the preprocessing processes are crucial. To guarantee a constant input size, the single-channel IR pictures and the three-channel RGB images are first downsized to a uniform square dimension - this differs according the model being evaluated.

The pictures are then transformed into a PyTorch Tensor format, which concurrently reduces the pixel intensity values from the initial $[0, 255]$ range to $[0.0, 1.0]$.

Normalization is the last important step. The transformation $T.Normalize((0.5, 0.5, 0.5), (0.5, 0.5, 0.5))$ is applied channel-wise to the RGB pictures, while a single-channel version $T.Normalize((0.5,), (0.5,))$ is used for the IR image. The pixel intensity distribution of both modalities is shifted from the $[0.0, 1.0]$ range to a $[-1.0, 1.0]$ range by this particular normalizing step.

5.4 Evaluation Metrics

Peak Signal-to-Noise Ratio (PSNR) Measures the reconstruction quality by quantifying the ratio between the maximum possible signal power (MAX^2) and the power of the corrupting noise (Mean Squared Error, MSE). Higher is better.

$$PSNR = 10 \log_{10} \left(\frac{MAX^2}{MSE} \right)$$

Structural Similarity Index Measure (SSIM) This metric evaluates the perceived quality based on the structural content of the images, combining luminance, contrast, and structural information. Closer to 1 is better.

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

Learned Perceptual Image Patch Similarity (LPIPS) The Learned Perceptual Image Patch Similarity (LPIPS) metric evaluates image similarity using deep feature embeddings extracted from pretrained networks such as AlexNet or VGG.

Instead of comparing pixels directly, LPIPS measures the distance between feature representations at multiple layers, which correlates more strongly with human perceptual judgments.

5.5 Model Architecture and Loss Functions

Pix2Pix

Architecture The initial conditional GAN formulation presented in the cGAN publication is followed by the Pix2Pix model, which acts as the study’s baseline. Its goal is to use paired supervision to learn a direct mapping from visible RGB pictures to their matching infrared equivalents. The model is composed of a discriminator that determines if an RGB–IR combination is synthetic or real, and a generator that forecasts the thermal picture from the visual input. Unlike unpaired GAN-based translation systems, Pix2Pix only uses paired training samples, therefore the quality and alignment of the dataset have a significant impact on the system’s performance.

The generator is constructed as an encoder-decoder made up of symmetric transposed-convolution upsampling layers after a sequence of downsampling convolutional layers. The network’s feature dimensionality grows from the input RGB channels to ever deeper feature maps as it descends through the encoder, ultimately arriving at a latent representation with 512 channels. After that, the decoder gradually increases spatial resolution to reassemble the infrared picture. A hyperbolic tangent activation function is used to normalize the single-channel infrared picture produced by the outermost layer such that the output falls within the range $[-1, 1]$.

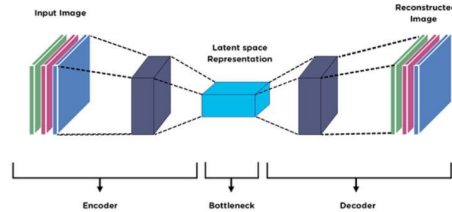


Fig. 3: Pix2Pix generator; encoder-decoder model

The PatchGAN design presented in the Pix2Pix study is followed by the discriminator. It generates a grid of patch-level authenticity predictions, each of which corresponds to a receptive field covering a tiny area of the input, rather

than producing a single real/fake conclusion for the full image. The model is encouraged by this structure to prioritize texture consistency and high-frequency realism over global picture categorization. After receiving the concatenation of the RGB input and either the produced or genuine infrared picture, the discriminator generates a single-channel map of authenticity scores after passing this four-channel input through a series of strided convolutions with increasing feature depth. The adversarial process expressly requires that the produced IR picture must be credible given the associated RGB image, not only realistic in isolation, by conditioning the discriminator on the visible input.

Loss The training objective follows the standard conditional GAN formulation. Let x denote a visible RGB image, y its corresponding infrared image, and $G(x)$ the generator output. The conditional adversarial loss is

$$\mathcal{L}_{cGAN}(G, D) = \mathbb{E}_{x,y}[\log D(x, y)] + \mathbb{E}_x[\log(1 - D(x, G(x)))].$$

The generator aims to minimize this loss while the discriminator maximizes it:

$$G^* = \arg \min_G \max_D \mathcal{L}_{cGAN}(G, D).$$

To enforce structural consistency between $G(x)$ and the ground-truth IR image, an ℓ_1 reconstruction loss is added:

$$\mathcal{L}_{L1}(G) = \mathbb{E}_{x,y} [\|y - G(x)\|_1].$$

The final Pix2Pix objective combines both terms:

$$G^* = \arg \min_G \max_D \left(\mathcal{L}_{cGAN}(G, D) + \lambda \mathcal{L}_{L1}(G) \right).$$

Hyperparameters Using a batch size of two, the baseline model (*Basic*) was trained for a total of twenty epochs. The Adam optimizer was used to optimize the generator and discriminator with a learning rate of 2×10^{-4} and β values of 0.5 and 0.999. The loss weighting factors for the generator were set to $\lambda_{L1} = 100$ and $\lambda_{SSIM} = 1.0$. The discriminator is a PatchGAN with 64 base filters and input channels = 4, whereas the generator architecture is a 4-level U-Net with input channels = 3, intermediate channel widths of 64, 128, 256, and 512s.

Hyperparameter & Architecture Study In order to identify the best configuration for visible-to-thermal translation, we compared two different pix2pix model setups and assessed the effects of picture resolution, augmentation procedures, and training stability methodologies.

Experimental Setup

- **Experiment A (Baseline):** Spatial resolution was given priority in this arrangement, which used consistent Center Cropping for training on 512×512 images. A limited batch size of two was employed due to memory limitations.

With an SSIM loss weight of $\lambda_{\text{SSIM}} = 1.0$, the learning rate was maintained at 2×10^{-4} .

- **Experiment B (Modified):** By lowering the resolution to 256×256 , this configuration sought to increase stability while enabling a batch size of six. We implemented Horizontal Flips and Random Cropping to avoid overfitting. We also used a linear learning rate decay, Label Smoothing (actual labels = 0.9), and raised the structural weight to $\lambda_{\text{SSIM}} = 5.0$.

Results & Analysis

The quantitative results, summarized in Table 1, reveal a significant performance divergence between the two approaches.

Table 1: Quantitative comparison of Baseline vs. Modified pix2pix configurations.

Model	Res.	Augment.	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$
Baseline (Exp A)	512 ²	Center	0.810	26.23	0.118
Modified (Exp B)	256 ²	Random	0.680	18.13	0.328

The Baseline configuration much outperformed the Modified version, defying the notion that larger batch sizes and augmentation would enhance generalization. The decline in Experiment B demonstrates that **resolution is paramount** for this work; the generator failed to recover high-frequency thermal features when the input size was reduced to 256×256 , as seen by the dramatic increase in LPIPS (0.328 vs. 0.118). Additionally, even though random augmentation usually helps with generalization, it made the job more difficult than could be successfully learnt in 20 epochs. In Experiment A, the model was able to concentrate on modality translation instead of spatial alignment because to Center Cropping’s consistent framing, which produced better structural fidelity.

ClawGAN

Architecture Inspired on CycleGAN, ClawGAN uses a dual-generator framework with a unique Claw-style U-Net++-like generator. The encoder gradually downsamples the input picture while extracting multi-scale feature representations using recurrent convolutional blocks in the encoder-decoder structure of the generator. The network may retain feature stability without batch statistics since each convolutional block comprises of a convolution layer, instance normalization, and ReLU activation. The architecture consists of two decoders: Decoder B, also known as the "claw branch," uses both the upsampled features and the intermediate outputs from Decoder A at corresponding levels, while Decoder A reconstructs the output using a traditional U-Net decoding path with skip connections from the encoder layers. The generator can record more complex feature correlations because to its dual-decoder architecture, which improves the translation of subtle face information. Convolution and hyperbolic tangent activation

are used to create the final output, which normalizes the created picture to the range $[-1, 1]$.

The discriminator is a PatchGAN that generates a grid of patch-level authenticity scores for the input picture using a sequence of convolutional layers with instance normalization and LeakyReLU activations. High-frequency detail preservation in the produced pictures is encouraged by this patch-based assessment.

Loss ClawGAN is trained using a combination of adversarial, reconstruction, and consistency-based losses. Let x denote a visible RGB image, y the corresponding thermal image, $G_{AB}(x)$ the generator mapping from visible to thermal, and D_B its discriminator. The adversarial loss for the generator is

$$\mathcal{L}_{\text{adv}}(G_{AB}, D_B) = \mathbb{E}_x [\log D_B(G_{AB}(x))],$$

while the discriminator minimizes

$$\mathcal{L}_{D_B} = \frac{1}{2} \mathbb{E}_y [\|D_B(y) - 1\|_2^2] + \frac{1}{2} \mathbb{E}_x [\|D_B(G_{AB}(x))\|_2^2].$$

To preserve structure and ensure semantic consistency, several auxiliary losses are used. The cycle-consistency loss is

$$\mathcal{L}_{\text{cyc}}(G_{AB}, G_{BA}) = \mathbb{E}_x [\|G_{BA}(G_{AB}(x)) - x\|_1],$$

supplemented by a synthesized loss $\mathcal{L}_{\text{syn}}$, an identity loss $\mathcal{L}_{\text{id}}$, and a feature-reconstruction loss $\mathcal{L}_{\text{fr}}$. The full ClawGAN generator objective is a weighted sum:

$$\mathcal{L}_{G_{AB}} = w_{\text{adv}} \mathcal{L}_{\text{adv}} + \lambda_{\text{syn}} \mathcal{L}_{\text{syn}} + \lambda_{\text{cyc}} \mathcal{L}_{\text{cyc}} + \lambda_{\text{id}} \mathcal{L}_{\text{id}} + \lambda_{\text{fr}} \mathcal{L}_{\text{fr}},$$

with an analogous objective for G_{BA} . These combined losses guide the dual-generator ClawGAN architecture to produce high-fidelity thermal translations while maintaining structural accuracy.

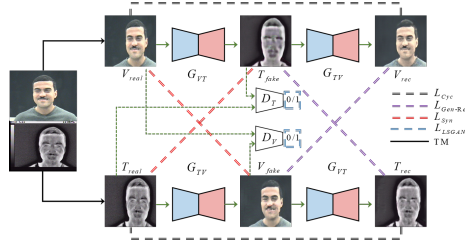


Fig. 4: ClawGAN model as per the paper[3]

Hyperparameters Across all experiments, several hyperparameters remained fixed. Each model was trained for 20 epochs using a batch size of 2 and an image resolution of 128×128 . The generator and discriminator were optimized using the Adam optimizer with a learning rate of 2×10^{-4} and β values of (0.5, 0.999). The network architectures remained constant throughout: the generator used a 4-level U-Net with base filter size 32, while the discriminator followed a Patch-GAN design with 64 base filters.

Hyperparameter Study Three experiments were conducted to evaluate how different loss-weighting schemes and dataset selections affected model performance and training stability. Table 2 summarizes the hyperparameters that varied between experiments. Parameters not listed remained identical across all runs.

Table 2: Hyperparameter variations across the three ClawGAN experiments.

Experiment	λ_{syn}	λ_{fr}	λ_{cycle}	λ_{id}	Adv. Weight
1	10.0	20.0	20.0	2.0	—
2	1.0	0.5	10.0	5.0	0.3
3	1.0	0.5	15.0	5.0	0.5

Experiment Analysis The IRIS thermal-visible dataset[15] was used to train the model for the first two investigations. The third research switched to the KAIST multispectral pedestrian dataset[5], which offers a wider and more varied training distribution, after resolving problems with loss stability and metric calculation.

Experiment 1 In the first experiment, BatchNorm-based convolutional blocks and baseline loss weights were used to train the model. The final epoch produced generator losses of $G_{AB} = 16.3895$, $G_{BA} = 12.8811$, and $G = 29.2706$. The total generator loss was much higher than intended, suggesting unstable optimization and a poor balance between identity, adversarial, and reconstruction constraints. By interfering with the translation mapping and causing scale inconsistencies and lighting distortions—a problem frequently seen in CycleGAN-like architectures—Batch Normalization made a detrimental contribution.

Figure 11 presents a representative visible-to-thermal translation example for Study 1, followed by the corresponding loss curves shown in Figure 6.



Fig. 5: Example visible-to-thermal translation result for Study 1.

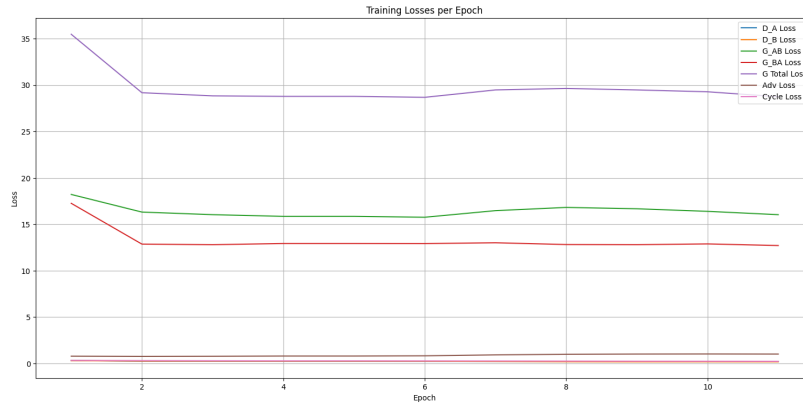


Fig. 6: Training loss curves for Study 1.

Experiment 2 The second experiment retained the same IRIS dataset but replaced BatchNorm with InstanceNorm and reduced several loss weights to prevent the generator from being excessively penalized. A dedicated adversarial weighting term was also introduced. Training produced much more stable dynamics, with final generator losses of $G_{AB} = 2.4462$, $G_{BA} = 2.0422$, and $G = 4.4884$, representing a substantial improvement over Study 1, particularly in total generator loss and cycle consistency. During this experiment, automatic evaluation metrics produced errors due to mismatched batch shapes and were corrected before Study 3. Figure 12 shows a sample visible-to-thermal output generated in Study 2. The training loss curves are plotted beneath it in Figure 13.



Fig. 7: Visible to Infrared to Visible (row 1) Infrared to Visible to Infrared (row 2)

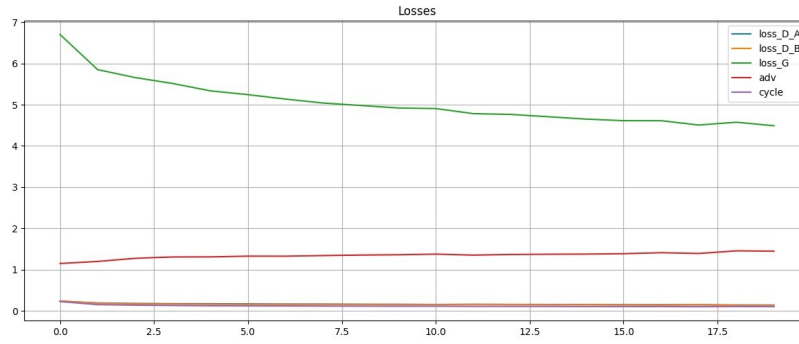


Fig. 8: Training loss curves for Study 2.

Experiment 3 The final experiment used the KAIST dataset, adopting the stabilized loss configuration from Study 2 while increasing the cycle-consistency weight and adversarial contribution. This study incorporated discriminator noise and restricted the dataset to a maximum of 20,000 samples for consistent epoch timing.

An example translation result produced in Study 3 is presented in Figure 9, with the corresponding loss plot displayed underneath in Figure 10.

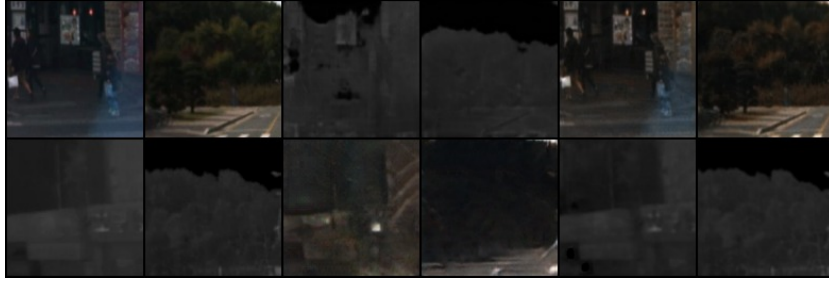


Fig. 9: Example visible-to-thermal translation result for Study 3.

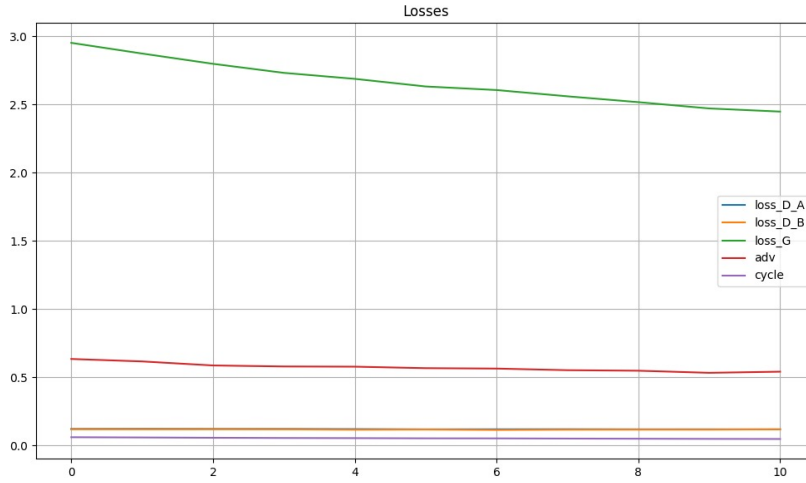


Fig. 10: Training loss curves for Study 3.

InfraGAN

Architecture InfraGAN maps visible RGB images x to their corresponding infrared images y using an encoder-decoder type generator, which allows the generator’s encoder-decoder architecture to enable the propagation of fine-grained spatial features from the input to the output. While the decoder upsamples using transposed convolutions and concatenates matching encoder feature maps at each level, the encoder gradually downsamples the input using convolutional layers with LeakyReLU activations. Using a tanh activation, the final output is normalized to lie into the interval $[-1, 1]$.

Two discriminators are used: a pixel-level discriminator that evaluates realism on a per-pixel basis using 1×1 convolutions and an encoder-decoder dis-

criminator that generates a spatial map of authenticity predictions. In order to enable the adversarial loss to condition on the visible input, both discriminators accept as input the concatenation of the RGB input image and either the produced or real infrared image. While the pixel discriminator uses small receptive field convolutions for fine-grained supervision, the encoder-decoder discriminator gradually downsamples and then upsamples feature maps, combining interim activations via skip connections.

Loss The generator is trained to minimize a combination of adversarial, L1, and structural similarity (SSIM) losses. Denoting $G(x)$ as the generator output and D_{ED} and D_{px} as the encoder-decoder and pixel discriminators respectively, the adversarial losses for the generator are

$$\begin{aligned}\mathcal{L}_{adv}^{ED}(G) &= \mathbb{E}_x [\text{GANLoss}(D_{ED}(x, G(x)), 1)], \\ \mathcal{L}_{adv}^{px}(G) &= \mathbb{E}_x [\text{GANLoss}(D_{px}(x, G(x)), 1)].\end{aligned}$$

with total adversarial loss $\mathcal{L}_{adv}(G) = \mathcal{L}_{adv}^{ED} + \mathcal{L}_{adv}^{px}$.

The L1 loss encourages pixel-wise similarity to the target infrared image:

$$\mathcal{L}_{L1}(G) = \mathbb{E}_{x,y} [\|G(x) - y\|_1],$$

and the SSIM loss promotes structural fidelity:

$$\mathcal{L}_{SSIM}(G) = \mathbb{E}_{x,y} [1 - \text{SSIM}(G(x), y)].$$

The total generator loss is a weighted sum of these components:

$$\mathcal{L}_G = \mathcal{L}_{adv} + \lambda_{L1}\mathcal{L}_{L1} + \lambda_{SSIM}\mathcal{L}_{SSIM}.$$

Discriminators are trained using standard LSGAN losses with mean squared error:

$$\mathcal{L}_D = \frac{1}{2}\mathbb{E}_{x,y} [\|D(x, y) - 1\|_2^2] + \frac{1}{2}\mathbb{E}_x [\|D(x, G(x))\|_2^2].$$

5.6 Hyperparameters

There was no specific hyperparameter search done for InfraGAN. Rather, the number of training samples was raised in a second experiment once the model showed stable learning behavior after an initial configuration was assessed on a part of the dataset. All other training parameters remained the same for both runs except for this modification. In particular, 256×256 photos, a batch size of 2, zero data-loading workers, and fixed weights of 100 for both the L1 and SSIM reconstruction terms were used to train both trials for 20 epochs. An image pool of size 50 was used in each case to smooth discriminator updates, consistent with the original InfraGAN training strategy. The only parameters that differed between the two experiments—namely the learning rates and the maximum number of training samples—are summarized in Table 3.

Table 3: Hyperparameters that differ between InfraGAN Experiment 1 and Experiment 2.

Parameter	Experiment 1	Experiment 2
Learning rate (Generator)	2×10^{-4}	1×10^{-5}
Learning rate (Discriminator)	2×10^{-6}	1×10^{-7}
Maximum samples used	10,000	20,000

Experiment 1



Fig. 11: Visible (left), Ground Truth (middle), InfraGAN (right)

Experiment 2



Fig. 12: Visible (left), Ground Truth (middle), InfraGAN (right)

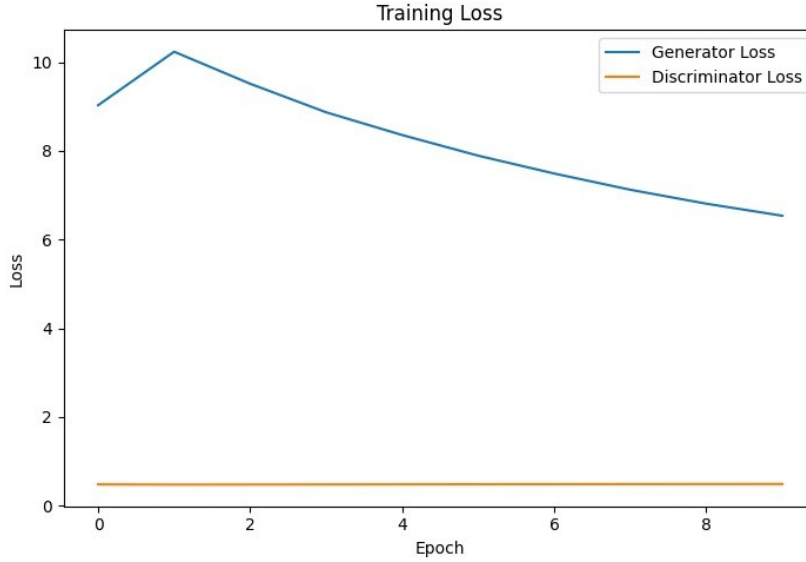


Fig. 13: Training loss curves for Experiment 2.

6 Quantitative Comparison

To evaluate the performance of the three models, we report Structural Similarity Index Measure (SSIM), Learned Perceptual Image Patch Similarity (LPIPS), and Peak Signal-to-Noise Ratio (PSNR). These metrics jointly quantify structural accuracy, perceptual realism, and pixel-level fidelity. Table 4 summarizes the results, with InfraGAN achieving the strongest performance overall.

Table 4: Quantitative comparison of Pix2Pix, ClawGAN, and InfraGAN using SSIM, LPIPS, and PSNR.

Model	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR (dB) $\uparrow$
Pix2Pix	0.810	0.118	26.23
ClawGAN	1.00	0.163	28.20
InfraGAN	0.9436	0.0935	35.9995

f0.810 **26.23** 0.118

7 Qualitative Comparison

A qualitative comparison between the three models is provided in Figure 14.

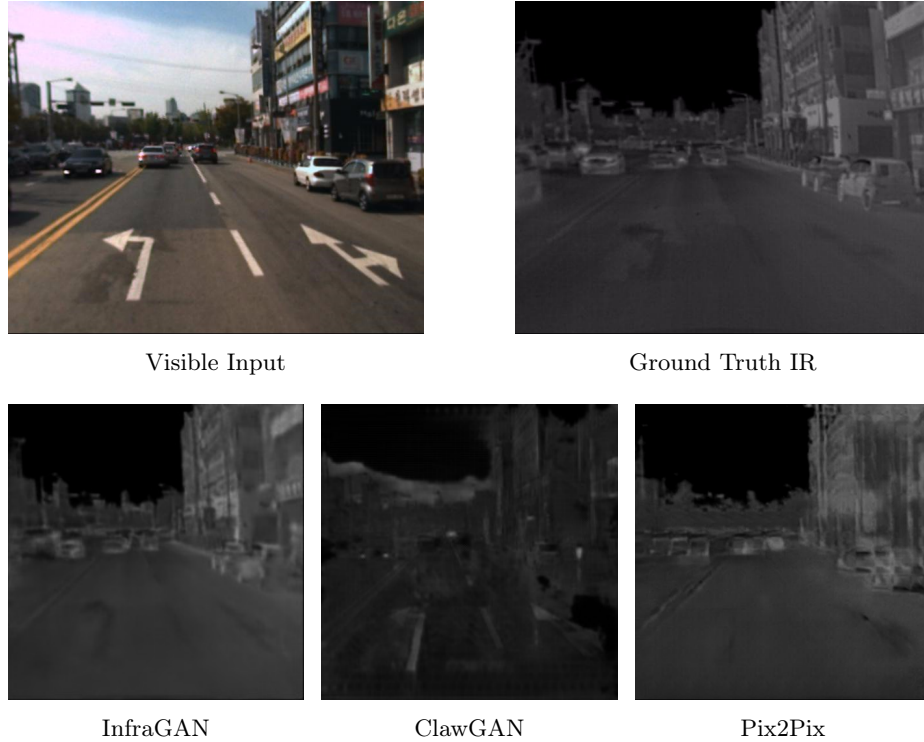


Fig. 14: Qualitative comparison of visible-to-thermal image translation. Top row: input visible image and corresponding ground truth IR image. Bottom row: generated IR images from InfraGAN, ClawGAN, and Pix2Pix.

8 Discussion

The three models’ qualitative and quantitative analyses demonstrate InfraGAN’s superior performance. InfraGAN routinely produced infrared images that closely resembled the ground reality, earning the top ratings across all evaluation measures. The efficiency of its architectural design is highlighted by its capacity to produce visually accurate outcomes with little hyperparameter modification. InfraGAN uses an encoder-decoder generator to transport fine-grained spatial information from the visible RGB input to the generated infrared output. While the decoder uses transposed convolutions to upsample by concatenating features from the appropriate encoder layers, the encoder gradually downsamples the input through convolutional layers with LeakyReLU activations.

In addition to adversarial and L1 components, a structural similarity (SSIM) term in the generator loss promotes the retention of structural content, which is especially advantageous for visible-to-infrared translation. Furthermore, both fine-grained and global supervision are provided by the dual-discriminator setup, which consists of an encoder-decoder discriminator that generates a spatial map

of authenticity predictions and a pixel-level discriminator that assesses realism on a per-pixel basis. By conditioning on the visible input, the discriminators enable the adversarial loss to direct the generator to generate realistic infrared images that match the input content.

Conversely, whereas ClawGAN demonstrated encouraging results on the IRIS facial dataset, it did not perform as well on the KAIST dataset. Because the reconstruction loss values were lower than the original construction objectives, the resulting infrared images were darker, blurrier, and less structurally cohesive. This implies that although ClawGAN’s dual-generator and claw-connection technique works well for some facial translation tasks, it has trouble generalizing to the more varied situations found in the KAIST dataset.

As a conditional GAN baseline, Pix2Pix yielded the least satisfying outcomes. The model’s shortcomings in capturing global structural consistency were evident in the output images, despite the patch-based discriminator’s ability to provide localized supervision. The benefits of InfraGAN’s dual-discriminator design and structural similarity loss were further highlighted by the outputs, which were typically hazy and lacked the fine-grained detail required for precise visible-to-infrared translation.

8.1 Future Works

The current generations created infrared images are intensity-based and grayscale. Adding temperature data or a predictive color map is a crucial development that goes beyond basic grayscale. In order to show the generated infrared image using a common thermal color palette (such as Ironbow or Rainbow), a post-processing module that maps the expected intensity values of the image to particular temperature ranges must be developed. For human analysis, this offers more representative and intuitive visuals.

The YOLO model would be applied to the resulting IR images, and the detector’s performance (such as Mean Average Precision, or mAP) may be used as a new, task-specific assessment metric. In addition to producing aesthetically beautiful images, high detection accuracy would show that the GAN is maintaining the crucial semantic knowledge needed for subsequent computer vision tasks.

We could consider substituting or enhancing the adversarial loss and the L1/L2 pixel-wise loss with a Perceptual Loss (VGG loss) to further enhance the structural fidelity of the synthesized images. The difference between the feature representations that a pre-trained network (such as VGG-19) extracts from the produced and actual images is measured by this loss.

9 Conclusion

In this study, we compared three models for translating visible-to-infrared images, and we found that InfraGAN consistently performs better than both ClawGAN and Pix2Pix in terms of quantitative measures and visual quality. Real-

istic and structurally authentic infrared images can be produced thanks to InfraGAN's dual discriminators and structural similarity loss. Pix2Pix's simpler patch-based cGAN framework resulted in less detailed and blurry outputs, while ClawGAN performed fairly well on facial datasets but struggled with the more varied KAIST scenarios.

Limitations, however, still do remain, especially with regard to the colorization of thermal imagery and the assessment of downstream activities. Future research will concentrate on producing colorized infrared outputs utilizing temperature-informed, as well as incorporating object detection metrics to evaluate the semantic accuracy of produced infrared images. Overall, this study emphasizes how crucial it is to use multi-scale discriminators and structural loss terms for high-quality cross-domain picture translation.

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